

Manikandan S

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EDUCATION

SASTRA Deemed University

Thanjore, Tamil Nadu, India

B.Tech, Computer Science & Engineering (IoT & Automation) – CGPA: 7.56 / 10.0

Aug. 2023 – May 2027

- **Relevant Coursework:** Machine Learning, Deep Learning, Data Structures & Algorithms, Operating Systems, Computer Networks, Database Management Systems
- **Research Interests:** Medical AI, Model Compression, Explainable AI, Efficient Deep Learning

RESEARCH PUBLICATION

Knowledge Distillation for Efficient and Lightweight Cancer Classification

Manikandan S, Aniruddh S B, Balaji B P — SAGE Publications (Under Review)

- Distilled a **7-model CNN teacher ensemble** (DenseNet121, InceptionV3, MobileNet, ResNet50, ResNet101, VGG16, Xception) into a lightweight **MobileNetV2 student** for histopathology cancer classification.
- Achieved **96.19% accuracy** on LC25000 with **98.41% model compression** (578.69 MB → 9.19 MB); cross-dataset validation on NCT-CRC-HE-100K via LUT-based stain normalization.

PROJECTS

Document AI Chatbot | *Python, LangChain, ChromaDB, HuggingFace, Gemini, Streamlit*

[GitHub](#)

- Built a **RAG-based PDF chatbot** using LangChain and ChromaDB for vector storage, enabling context-aware question answering over user-uploaded documents.
- Integrated **HuggingFace sentence-transformer embeddings** for semantic chunking and **Google Gemini** as the generative backbone; deployed end-to-end on **Streamlit Community Cloud**.
- Implemented persistent vector store with automatic document ingestion pipeline supporting multi-page PDFs and real-time streaming responses.

Salon Slot Booking System | *Node.js, Express.js, SQLite, REST API, HTML/CSS/JS*

[GitHub](#)

- Developed a full-stack **salon appointment booking web app** with Node.js/Express backend and SQLite for persistent slot management across sessions.
- Designed **RESTful API endpoints** for creating, viewing, and cancelling bookings with conflict detection to prevent double-booking of time slots.
- Built a responsive frontend with vanilla HTML/CSS/JS and an admin dashboard for real-time slot availability management.

Hybrid Self-Evolving Neural Network | *Python, TensorFlow, Flask, AutoML*

[GitHub](#)

- Designed an **AutoML system** that generates and optimises deep learning architectures from a user prompt using an evolutionary search pipeline with mutation and crossover mechanisms.
- Automated input shape correction and architecture validation across **image classification, sequence, and regression tasks**; exposed training workflows via RESTful API endpoints.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, Java

AI / ML: NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Transformers, ML Pipelines, SHAP, RAG, LangChain

Backend: FastAPI, Flask, REST APIs, Node.js, SQL, MongoDB, Postman

Systems & Tools: Linux, Bash, Docker, Git/GitHub, Networking Basics, Self-hosted Server Administration